

Instructional Design Document

Goal Statement

In today’s healthcare environment, nurses and clinical staff rely heavily on digital tools like electronic health records, email, and hospital systems to deliver patient care. Unfortunately, this also makes them prime targets for phishing and other cyber threats. According to the U.S. Department of Health and Human Services (2024), nearly 79% of healthcare data breaches were caused by hacking or IT incidents—most often phishing and social engineering attacks. These breaches can lead to serious consequences, including financial loss, compromised patient privacy, and disruptions to clinical operations. This training is designed to help frontline healthcare staff recognize and respond to common digital threats. The goal is to build practical awareness and confidence so learners can protect sensitive patient information and take appropriate action when they encounter suspicious messages. By the end of the course, participants will be able to:

- Identify common phishing tactics
- Apply secure communication practices
- Follow hospital protocols to report potential threats

The course uses realistic simulations and a downloadable checklist to support learning and reinforce best practices on the job.

Reference: U.S. Department of Health and Human Services (2024). Office for Civil Rights Breach Report.

Learner Analysis

This training is designed for nurses and clinical support staff working in hospital settings. These professionals are highly skilled in patient care but vary widely in their comfort with technology and cybersecurity concepts. Some have encountered suspicious emails before but aren’t sure how to respond, while others may not recognize phishing attempts at all.

Learners operate in fast-paced environments with limited time for training. They need instruction that is clear, concise, and immediately applicable to their daily routines. To support this, the course includes a mix of visual, auditory, and interactive elements—such as simulations, infographics, and printable job aids. Accessibility features like closed captions, high-contrast visuals, and zoom functionality are built in to ensure usability across roles and abilities. To better understand learner needs, we developed three representative personas. These profiles helped guide design decisions around pacing, media selection, and support tools. The table below summarizes key characteristics and corresponding design implications:

Persona-Based Learner Analysis

Attribute	Maria (RN, age 29) & Description	Design Implication	James (Support Staff, age 46) & Description	Design Implication	Aisha (NP, age 35) & Description	Design Implication
Learner Profile	Tech-comfortable, multitasks across hospital systems; quick to adapt	Use interactive simulations and scenario-based learning to maintain engagement	Tech-anxious, prefers simplicity; limited exposure to cybersecurity	Provide step-by-step guides, printable checklists, and low-pressure practice	Tech-savvy, confident, self-directed; seeks depth and autonomy	Include optional deep dives, peer-led case studies, and leadership prompts
Entry Skills	Familiar with hospital email and messaging tools; some phishing awareness	Begin with quick refreshers, then move to complex decision-making scenarios	Basic email literacy; unfamiliar with phishing tactics or reporting	Scaffold foundational concepts with microlearning and visual aids	Advanced digital fluency; understands secure platforms and protocols	Skip basics, offer advanced simulations and mentoring opportunities
Motivation	High—driven by patient safety and accountability	Frame training as protecting patients and supporting hospital mission	Moderate—motivated by compliance, but fears making mistakes	Emphasize psychological safety and normalize error reporting	High—motivated to lead and mentor others	Position training as a leadership opportunity with peer reinforcement
Learning Preferences	Prefers interactive visuals, simulations, and real-world scenarios	Use drag-and-drop, branching scenarios, and visual infographics	Prefers simple, structured formats like checklists and guided walkthroughs	Include printable job aids, linear navigation, and clear feedback	Prefers autonomy, case-based learning, and reflection opportunities	Offer flexible navigation, enrichment modules, and peer discussion prompts
Accessibility Needs	Needs mobile-friendly access, closed captions, and visual clarity	Ensure mobile optimization and captioning for all media	Needs high-contrast visuals, large text, and printable resources	Use high-contrast design and offer offline resources	Needs flexible access and downloadable reference materials	Provide downloadable tools and optional accessibility toggles

Time Constraints	Learns during shift breaks; prefers short, focused modules	Chunk content into ≤5-minute segments with progress tracking	Has unpredictable workflow; needs pause/resume functionality	Allow flexible pacing and bookmarking features	Can complete full module in one sitting; prefers structured flow	Design seamless module flow with optional checkpoints for reflection
Attitudes Toward Cybersecurity	Sees it as part of patient safety but unsure how to act	Use peer-led scenarios to build confidence and normalize action	Hesitant to report; fears judgment or reprimand	Reinforce psychological safety and provide anonymous practice simulations	Believes in shared accountability and wants to model best practices	Include leadership framing and peer challenge activities
Cultural Dynamics	Influenced by peer norms; responds well to team-based reinforcement	Include peer huddle prompts and team-based reflection tools	Responds best to manager encouragement and clear expectations	Provide manager toolkits and follow-up discussion guides	Comfortable speaking up; values collaboration and feedback	Embed peer debriefs and leadership challenges to foster engagement

Performance Context Analysis

Nurses and clinical support staff will apply the skills from this training directly in their day-to-day hospital work environments. These settings are fast-paced, high-pressure, and heavily reliant on digital communication. Staff routinely access electronic health records (EHRs), respond to emails, and manage patient data, often while multitasking or under time constraints.

Cyber threats like phishing do not arrive with warning labels. They show up in inboxes during busy shifts, disguised as routine messages. A single click on a malicious link can compromise patient privacy, disrupt clinical operations, and trigger costly investigations. That's why this training focuses on helping learners recognize threats in real time and respond confidently, even when distracted or under stress.

Some staff may hesitate to report suspicious messages for fear of being wrong or judged. Others may not know who to contact or what steps to follow. The course addresses these barriers by normalizing reporting, reinforcing shared responsibility, and walking learners through hospital-approved protocols.

Staff may use shared workstations, mobile devices, or tablets. The training is designed to be mobile-friendly and accessible across platforms, with short segments that can be completed between tasks or during downtime.

Key Contextual Factors:

- **Digital workflow:** Frequent use of EHRs, email, and internal systems
- **Time pressure:** Limited windows for training and decision-making
- **Social hesitation:** Fear of reporting incorrectly or appearing uninformed
- **Device variability:** Mix of desktop, mobile, and shared access
- **Cultural emphasis:** Patient safety and professional accountability are strong motivators

By designing these realities, the course ensures that learners can apply cyber hygiene practices confidently and consistently, without disrupting their clinical responsibilities.

Learning Environment Context

This course will be delivered as a 30-minute asynchronous eLearning module hosted on the hospital's secure network. It is designed to be completed in one sitting but can also be paused and resumed to accommodate shift-based schedules. Learners will access the course using hospital-issued computers or tablets, ensuring compatibility with internal systems and security protocols.

The learning experience is entirely online and self-paced, allowing staff to complete it during downtime or between clinical responsibilities. The course includes interactive simulations that mirror real-world threat scenarios in a hospital setting, helping learners build confidence in recognizing and responding to phishing attempts.

To ensure accessibility and ease of use across roles and devices, the course includes the following features:

- **Closed captions** for all audio and video content
- **Zoom in/out** functionality for visual clarity
- **High-contrast color options** to support visual accessibility
- **Infotips** that provide additional guidance when learners hover over icons or visuals
- **Mobile-responsive layout** for flexible access on tablets or shared workstations
- **Downloadable one-page checklist** that learners can reference during and after the course

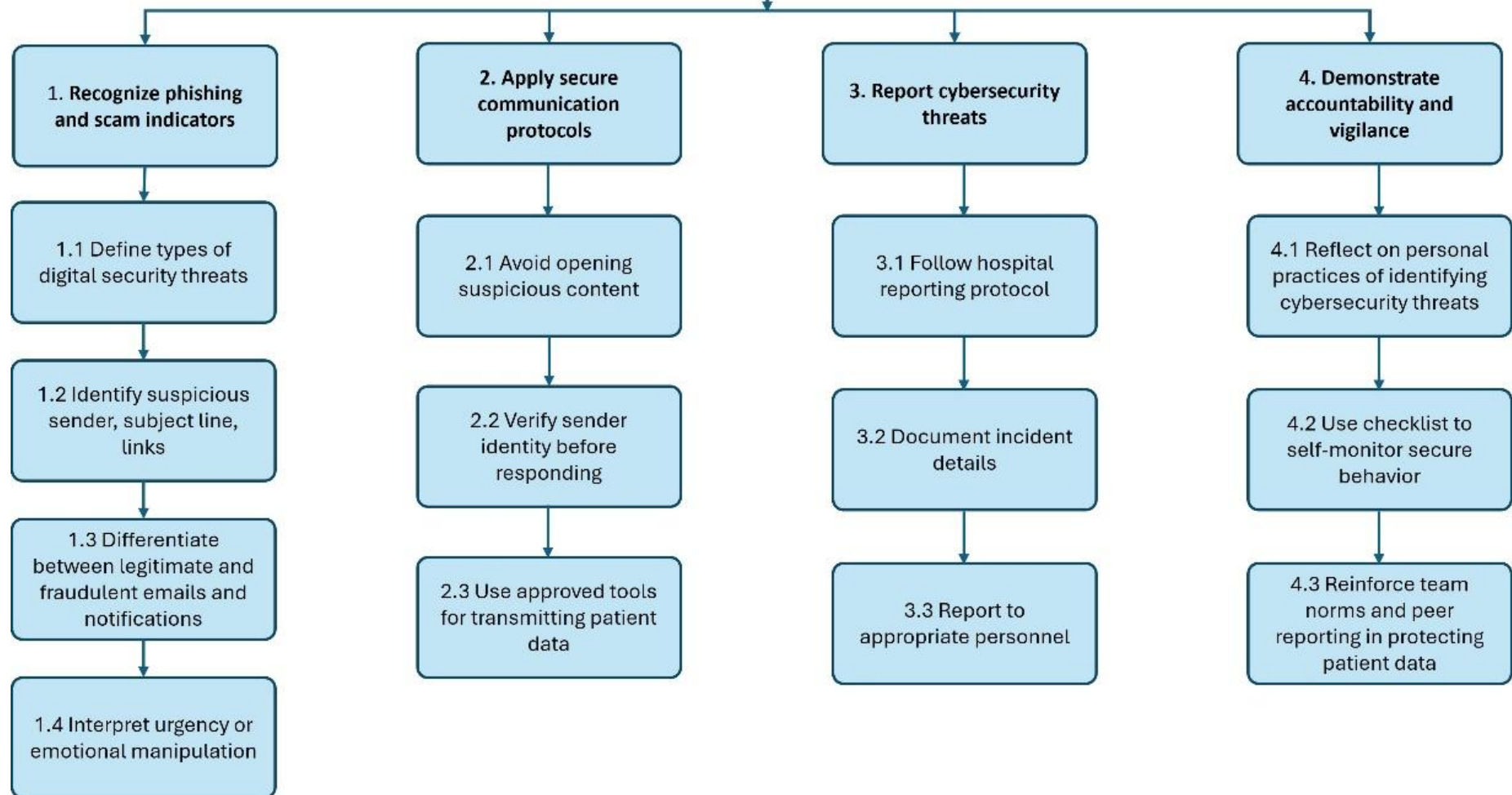
The course content is adapted from trusted sources, including the Cybersecurity and Infrastructure Security Agency (CISA), and is tailored to reflect hospital-specific communication tools and reporting protocols.

By designing for real-world constraints—such as limited time, shared devices, and varying levels of tech comfort—the course ensures that all learners can engage meaningfully with the content and apply it immediately on the job.

Task Analysis

To design effective instruction, it's essential to understand what learners must do to perform successfully in their real-world roles. Task analysis breaks down the overall performance goal into smaller, observable actions and decisions. In this training, the goal is to help healthcare staff recognize, respond to, and report phishing threats confidently and accurately. The analysis below outlines the key tasks and subtasks required to achieve this goal, based on the instructional context and learner needs described earlier.

Task Analysis



Task ID	Task/Subtask Description	Cognitive Process	Assessment Strategy
1	Recognize phishing and scam indicators	Whole-task awareness: Learners must scan digital messages and apply pattern recognition to detect threats.	Scenario-based prompt: Learners view a simulated inbox and identify suspicious messages.
1.1	Define types of digital security threats	Verbal information recall: Learners retrieve and label key threat categories (e.g., phishing, malware, ransomware) and understand their basic characteristics.	Matching activity + guided infographic: Learners match threat types to definitions and explore examples via an interactive visual.
1.2	Identify suspicious sender, subject line, links	Discrimination: Learners distinguish between safe and unsafe message elements based on visual and textual cues.	Drag-and-drop activity: Learners sort message components into “safe” vs. “suspicious.”
1.3	Differentiate between legitimate and fraudulent emails and notifications	Concept classification: Learners categorize examples based on structure, tone, and sender credibility.	Interactive infographic + quiz: Learners explore phishing anatomy and answer classification questions.
1.4	Interpret urgency or emotional manipulation	Pattern recognition: Learners detect psychological tactics used in phishing (e.g., fear, urgency, reward).	Scenario walkthrough: Learners analyze message tone and intent in branching examples.
2	Apply secure communication protocols	Rule-based application: Learners apply hospital-approved procedures to real-world messaging scenarios.	Simulation feedback: Learners receive immediate feedback on secure vs. insecure choices.
2.1	Avoid opening suspicious content	Inhibition + decision-making: Learners resist impulsive actions and make deliberate choices.	Simulation feedback: Learners receive corrective feedback when risky actions are selected.
2.2	Verify sender identity before responding	Rule application: Learners follow a verification checklist before replying to messages.	Simulation walkthrough: Learners practice verifying senders using hospital tools.

2.3	Use approved tools for transmitting patient data	Procedure execution: Learners follow step-by-step instructions to send data securely.	Case study + checklist: Learners apply secure messaging protocols in a realistic scenario.
3	Report cybersecurity threats	Inhibition + decision-making: Learners resist impulsive actions and make deliberate choices.	Simulation feedback: Learners receive corrective feedback when risky actions are selected.
3.1	Follow hospital reporting protocol	Procedure execution: Learners complete a reporting form or follow a documented process.	Final quiz: Learners demonstrate correct steps in a simulated reporting workflow.
3.2	Document incident details (sender, time, content)	Information recall: Learners retrieve and record relevant information from the message.	Job aid usage: Learners use a downloadable checklist to guide documentation.
3.3	Report to appropriate personnel	Decision-making: Learners choose the correct contact based on role and incident type.	Scenario-based branching: Learners select who to notify in different threat situations.
4	Demonstrate accountability and vigilance	Attitudinal shift + strategic behavior: Learners internalize cyber hygiene practices and apply them proactively.	Post-course survey + peer reinforcement: Learners reflect on behavior and share practices.
4.1	Reflect on personal practices of identifying cybersecurity threats	Metacognition: Learners evaluate their own messaging habits and risk awareness.	Self-check prompt: Learners respond to reflective questions about their behavior.
4.2	Use checklist to self-monitor secure behavior	Strategy use: Learners apply a structured tool to reinforce safe messaging habits.	Job aid usage: Learners download and use a checklist during and after training.
4.3	Reinforce team norms and peer reporting in protecting patient data	Social modeling: Learners promote cyber hygiene within their teams and normalize reporting.	Manager toolkit + peer discussion: Managers facilitate team huddles using provided resources.

Learning Objectives

To ensure healthcare staff can confidently recognize and respond to phishing threats, the course is built around four core learning objectives. These objectives reflect the real-world tasks learners must perform and are designed using the ABCD model (Audience, Behavior, Condition, Degree). Each objective is grounded in the performance gaps identified earlier and supports the development of both technical skills and professional accountability. The objectives guide the design of instructional activities, assessments, and performance support tools throughout the course.

Objective	Audience (A)	Behavior (B)	Condition (C)	Degree (D)	Learning Domain
1	Nurses and clinical support staff	Identify phishing indicators	Given simulated email scenarios	With 80% accuracy	Intellectual Skills – Discrimination
2	Nurses and clinical support staff	Apply secure communication protocols	During interactive simulations	With 80% accuracy	Intellectual Skills – Rule Application
3	Nurses and clinical support staff	Report cybersecurity threats	Using hospital protocol	With 80% accuracy	Intellectual Skills – Problem Solving
4	Nurses and clinical support staff	Demonstrate accountability and vigilance	Using a self-checking checklist	Independently and consistently	Attitude – Professional Responsibility

1. Identify phishing indicators

Learners will be able to:

- Define key healthcare cybersecurity terms using a glossary.
- Recognize suspicious sender addresses, URLs, and message elements.
- Detect urgency, threats, and emotional manipulation in digital communication.
- Differentiate between legitimate and fraudulent formats (e.g., email, text, notifications).

2. Apply secure communication protocols

Learners will be able to:

- Select appropriate tools for transmitting patient data (e.g., encrypted messaging platforms).
- Avoid insecure practices such as texting PHI or using personal email accounts.
- Follow procedural steps using a secure messaging checklist provided in the course.

3. Report cybersecurity threats

Learners will be able to:

- Access and navigate the hospital's internal reporting system.
- Document incident details accurately, including sender, time, and message content.
- Escalate threats to appropriate personnel following hospital protocol.

4. Demonstrate accountability and vigilance

Learners will be able to:

- Use a printable checklist to self-monitor cyber hygiene practices.
- Reflect on their personal responsibility in safeguarding patient information.
- Participate in peer debriefs or manager-led huddles to reinforce shared accountability.

Instructional Strategies

This course uses a blended instructional strategy that combines scenario-based learning, interactive media, and performance support tools. The design reflects the realities of hospital environments—limited time, varied tech comfort, and high stakes for patient data security. Each strategy is mapped to specific learner needs and cognitive demands, ensuring both engagement and transfer to practice.

Pre-Instructional Activities

Purpose: Prepare learners by activating prior knowledge, establishing relevance, and reducing anxiety.

Strategy	Description
Animated Intro Video	Introduces the importance of cybersecurity in healthcare using real-world breach statistics and consequences.
Self-Reflection Prompt	Learners answer: “Have you ever received a suspicious message at work? What did you do?” to surface prior experience.
Interactive Glossary	Learners explore key terms (e.g., phishing, spoofing, ransomware) before diving into scenarios.
Initiating Pre-Test (Optional)	A short 5-question quiz gauges baseline awareness and primes learners for upcoming modules.

Presentation Activities

Purpose: Deliver core content using multimodal, scenario-based instruction.

Strategy	Description
Branching Scenarios	Learners navigate simulated inboxes and make decisions about suspicious messages.
Infographics & Visual Comparisons	Show legitimate vs. fraudulent formats across email, text, and app notifications.
Case Studies	Realistic vignettes illustrate consequences of insecure communication and reporting failures.
System Walkthroughs	Learners explore hospital reporting tools and secure messaging platforms in guided simulations.

Practice Activities

Purpose: Reinforce learning through interactive, feedback-rich experiences.

Strategy	Description
Drag-and-Drop Sorting	Learners classify message components as “safe” or “suspicious.”
Interactive Checklist Walkthrough	Learners apply secure messaging steps using a downloadable checklist.
Peer Debrief Simulation	Learners engage in mock team huddles to reinforce accountability and shared norms.
Reflection Prompts	Learners respond to questions like “Would I report this message?” to build metacognitive awareness.

Assessment Activities

Purpose: Measure learning outcomes, provide feedback, and support remediation.

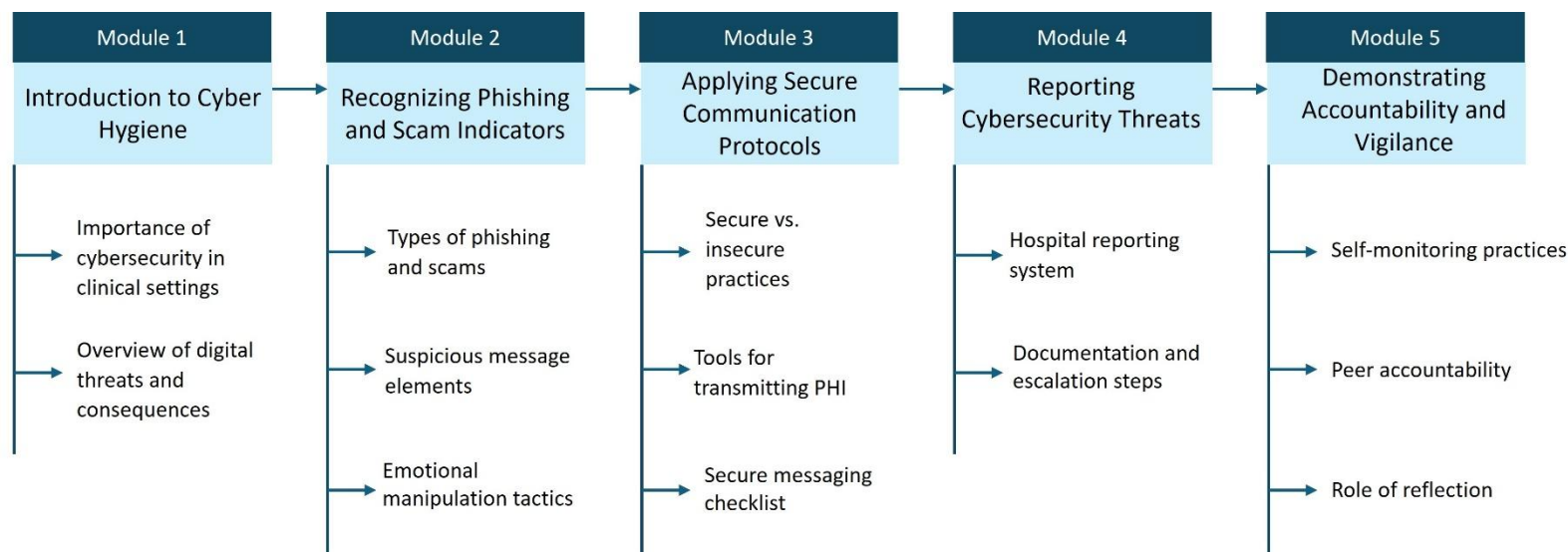
Strategy	Description
Formative Assessments	Embedded throughout modules (e.g., quizzes, scenario decisions) with immediate feedback.
Summative Assessment	Final scenario-based quiz in Module 4 evaluates secure messaging and reporting accuracy.
Remediation Opportunities	Learners can revisit modules and retry assessments. Incorrect responses trigger hints and explanations.
Job Aid Integration	Printable checklists and reporting guides support performance beyond the course.
Post-Course Survey	Captures learner confidence, behavioral intentions, and feedback for continuous improvement.

Cross-Cutting Design Features

- **Chunked Content:** Modules are broken into 5–7 minute segments for flexible completion.
- **Multimodal Delivery:** Combines text, audio, visuals, and interactivity to support diverse learning preferences.
- **Accessibility:** Closed captions, high-contrast visuals, zoom functionality, and mobile responsiveness.
- **Performance Support Tools:** Job aids, checklists, and printable resources embedded throughout and available post-training.

Instructional Content Map

This content map outlines the structure of the course, showing how each module aligns with specific tasks and subtasks. It includes the content focus, media formats, and embedded assessments or performance support tools.



Module	Title	Content Focus	Media/Activities	Mayer's Principles	Assessment & Support Tools
Mod. 1	Introduction to Cyber Hygiene	- Importance of cybersecurity in clinical settings - Overview of digital threats and consequences	- Animated explainer video - world incident case vignette - Interactive glossary	- Coherence - Segmenting	- Knowledge check (5-item quiz) - Downloadable glossary
Mod. 2	Recognizing Phishing and Scam Indicators	- Types of phishing and scams - Suspicious message elements - Emotional manipulation tactics	- Drag-and-drop sorting activity - Interactive email dissection - Branching scenario: inbox simulation	- Signaling - Spatial Contiguity	- Scenario-based quiz - Visual comparison infographic

Mod. 3	Applying Secure Communication Protocols	- Secure vs. insecure practices - Tools for transmitting PHI - Secure messaging checklist	- Simulation: secure messaging workflow - Case study: texting PHI error - Interactive checklist walkthrough	Modality and Redundancy	- Simulation feedback - Downloadable secure messaging checklist
Mod. 4	Reporting Cybersecurity Threats	- Hospital reporting system - Documentation and escalation steps	- Guided system walkthrough - Branching simulation: reporting decision tree - Job aid integration	Temporal Contiguity, Segmenting	- Final quiz (scenario-based) - Printable reporting job aid
Mod. 5	Demonstrating Accountability and Vigilance	- Self-monitoring practices - Peer accountability - Role of reflection	- Reflection prompts - Peer debrief guide - Manager-led huddle toolkit	Generative Processing	- Post-course survey - Printable self-monitoring checklist